

One screen, four steps

Production-order create/edit, from a one-line prompt to a merged pull request — basic design, database design, detailed design and implementation, each recorded and transcribed.

All four presentations in one file, **in the order the work ran**. Each part keeps its own title slide, and its step numbers match its edited video.

Recorded 2026-09-18 · Claude Code (Opus 5) in the JetBrains Rider terminal, auto mode on.

Basic design — 3:49 video

Database design — 3:59 video

Detailed design — 4:00 video

Implementation — 5:59 video

17:48 of video · 64 slides

Four parts, in the order the work ran

Each part is a self-contained presentation with its own recorded video. The parts connect: every one ends by naming what the next picks up.

PART	STEP	VIDEO	PAGES
Part 1	Basic design Requirements brief, decision log and a 339-line basic design	3:49	4–18
Part 2	Database design The schema, eight decisions and a new requirement	3:59	19–33
Part 3	Detailed design Four design documents and a published mockup	4:00	34–48
Part 4	Implementation The screen itself: 133 tests, two pull requests	5:59	49–63

HOW TO READ A PART

A title slide, then *the session at a glance*, then *the walkthrough* — a grid of the video's steps — then one slide per step, and finally what the step shows and where it leaves the project.

READING THE SLIDES

Amber marks what the human did; **teal** marks what the agent did. Every step slide carries a VIDEO mm:ss badge pointing into that part's edited video.

Eighty-seven minutes of recording, about an hour of agent work

Thirteen typed prompts and four question forms across the four steps. Everything else was the agent working through the harness in `ai/`.

PART	RECORDING	PROMPTS	AGENT TIME	WHAT IT PRODUCED
1 Basic design	9:59	2	6 m 16 s	Requirements brief, decision log, BD-001 (339 lines), plan, status, evidence
2 Database design	10:44	3	4 m 19 s	DB-002 (216 lines), DEC-013–DEC-020, REQ-019, BD-001 revision 3
3 Detailed design	15:37	2	10 m 23 s	Four DD documents (1,314 lines) and a mockup published as a private artifact
4 Implementation	50:44	6	~40 m	The screen: backend, database, frontend, 133 tests, two pull requests

WHAT STAYED CONSTANT

Every step read the harness before writing, recorded each decision with its options and rationale, and stopped at a gate it had been given in writing — plan approval, publishing the mockup, merging.

WHAT THE HUMAN DID

Chose between options, and twice asked a question that changed the session: "*are there any open questions left?*" and "*there are still 2 templates not output*". Both surfaced things the agent had quietly settled on its own.

Screen A: basic design

Two one-line prompts and four answers, and Claude Code produced the requirements brief, the decision log, the plan, and a 339-line basic design for the production-order create/edit screen.

Presented alongside `ScreenA_BD_edited.mp4` (3 min 49 s). Step numbers on the slides match the captions in the video.
Recorded 2026-09-18 · Claude Code (Opus 5) in the JetBrains Rider terminal, auto mode on.

`brief.md`

`decisions.md`

`BD-001-...-create-edit.md`

`plan.md`

`status.md`

`evidence.md`

Ten minutes of recording, six of agent work

The human wrote two prompts and answered one form. Everything else was the agent working through the harness.

2

prompts typed

both one line long

4

decisions answered

in a single form

6 m 16 s

agent working time

4 m 39 s + 1 m 37 s

6

files created

plus 3 index/docs updates

The human

- Asked for Screen A "as per what we planned".
- Noticed open questions were still sitting in BD-001 and asked for them.
- Chose an answer for each of the four questions. For the order-number format they **overrode the recommendation**.

The agent, guided by the harness

- Picked the workflow, skills, template and checklist without being told.
- Reused the business answers already agreed (DEC-001 to DEC-008) rather than asking again.
- Wrote every artifact, recorded every answer, and stopped at the plan-approval gate.

12 steps, one video

Each step is one or more clips in the edited video. **Amber: the human acts.** **Teal: the agent works and produces output.**

STEP 0100:00 Prompt 1 Start Screen A as planned	STEP 0200:14 Reads the harness Workflow, skills, template, checklist	STEP 0300:37 brief.md 70 lines, REQ-010 to REQ-018	STEP 0400:51 decisions.md 261 lines, DEC-001 to DEC-009
STEP 0501:07 BD-001 339 lines of basic design	STEP 0601:25 plan · status · evidence And the docs indexes	STEP 0701:41 Turn 1 summary One new question: DEC-009	STEP 0802:03 Prompt 2 "Still open decisions, right?"
STEP 0902:15 Decision form Four questions, one form	STEP 1002:48 Recording the answers DEC-009 to DEC-012	STEP 1103:16 Turn 2 summary What is left for DB and DD	STEP 1203:38 Review BD-001, item by item

Times are positions in `ScreenA_BD_edited.mp4` . Long stretches of agent work are jump-cut to their result; nothing is sped up.

One line starts the whole flow

THE PROMPT

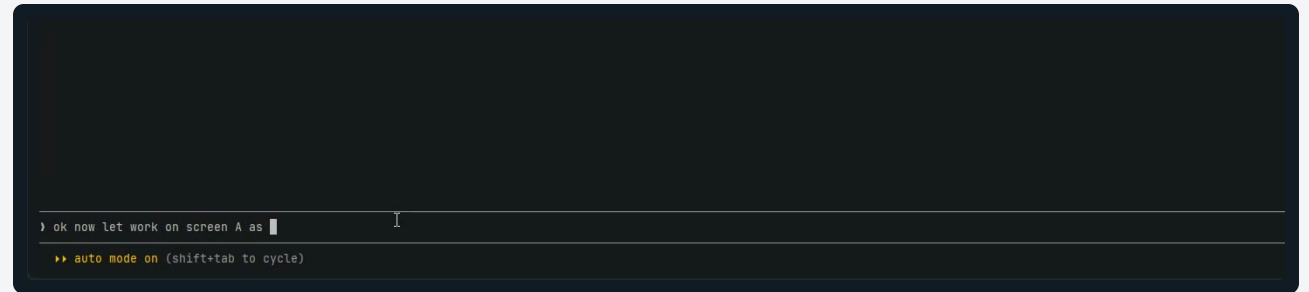
OK, now let's work on Screen A as per what we planned

No files, templates or steps are named. Claude works them out from the repository harness:

AGENTS.md → ai/project.md → ai/workflows → ai/skills →
ai/templates → ai/checklists

OUTPUT

No output yet. The prompt only sets the goal; the harness decides the route (step 2).



Video 00:07. The prompt being typed into Claude Code.

Read first, write later

WHAT CLAUDE READ	WHAT IT GOT FROM IT
AGENTS.md, ai/project.md, work-items/	Project state; Screen A is WI-002 (production-order create/edit)
feature-delivery workflow, requirements + basic-design skills	The route: brief first, then basic design
BD template, documentation + frontend rules, design checklist	The structure BD-001 must follow and the bar it must meet
Existing REQ / FN / SCR IDs, WI-001 roles, backend auth	Numbering starts at REQ-010; roles are Admin and Operator
Example Excel BD workbook	Table conventions for items, validation and events



OUTPUT

"Inputs gathered. Writing the WI-002 work item (brief, decisions, plan, status, evidence) and BD-001 against the template now." **About 80 seconds of read-only commands before a single file is written.**

```
ok now let work on screen A as per what we planned

Listed 3 directories, ran 3 shell commands

• Starting Screen A from the requirements brief and basic design. I'm now reading the BD template to make sure the new BD follows it.

Searched for 1 pattern, read 1 file, listed 1 directory, ran 3 shell commands

• Still gathering inputs - checking the example BD workbook's validation/event conventions before writing the brief and BD.

Ran 1 shell command

• Inputs gathered. Writing the WI-002 work item (brief, decisions, plan, status, evidence) and BD-001 against the template now, reusing the business answers you already agreed.
```

Video 00:31. Read-only commands, collapsed by Claude Code.

The work item comes first

STEP 3 — WORK-ITEMS/WI-002/BRIEF.MD · 70 LINES

- **Objective:** show the full lifecycle (requirements → BD → DB → DD → code → test → PR) on one small but realistic screen.
- **Success metrics:** full traceability, business rules proven by automated tests, a working demo in local Docker Compose.
- Requirements **REQ-010 to REQ-018**, each with success and failure criteria.
- **Out of scope:** the list screen and dashboard, a product-catalog UI, deleting orders, audit history.

```
8
9 Revision 1, 2026-09-18. The business answers already agreed with you are recorded in 'decisions.md' DEC-001-DEC-008.
...
74 lines
• Update(D:\Work\AI\WMS\work-items\WI-002\brief.md)
  - Removed 1 line
  59 - Production-order list/search (Screen B, separate work item) and dashboard (Screen C).
  60 - Product catalog create/edit UI - products are seeded reference data only (DEC-005).
  61 - Deleting production orders - 'Cancelled' is the only way to retire an order.
  62 - Editing any field of a 'Completed' or 'Cancelled' order other than what DEC-004 allows (see BD-001 for the exact per-status editability).
  63 - Audit history / change log beyond created/updated timestamps.
  64 - Finer-grained permissions than "Admin or Operator" (WI-001 DEC-015 stays open for other screens).
  65
• Now the decision log.
```

Video 00:44. The brief lands in the work item.

STEP 4 — DECISIONS.MD · 261 LINES

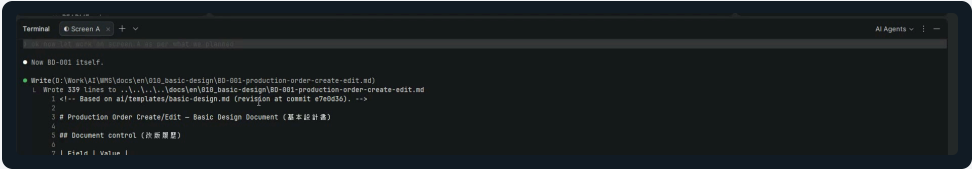
ID	DECISION	OUTCOME
DEC-001	Who may create/edit	Admin and Operator
DEC-003	Status workflow	Draft → InProgress → Completed; Cancelled from either
DEC-004	Edit once not Draft	Product and quantity locked
DEC-006	Validation	Qty > 0, due date ≥ today, notes ≤ 500
DEC-007	Locked field sent anyway	Reject the whole request
DEC-009	Overdue orders	Proposed — waiting for the user

Eight answers the user had **already agreed** go straight in. Only one new question is raised.

BD-001: the screen, specified

docs/en/010_basic-design/BD-001-production-order-create-edit.md · 339 lines · built from ai/templates/basic-design.md

SECTION	WHAT IT PINS DOWN
System overview	Every requirement mapped to the sections that cover it
Architecture	React + .NET 10 API + PostgreSQL 17, WI-001 auth; no new boundary
Function list	FN-001 Create ... FN-008 Enforce authentication/authorization
Status diagram	Draft → InProgress → Completed, Cancelled; two terminal states
SCR-001 detail	Routes, API paths, roles, page titles, PC and SP layouts
Mockups	ASCII PC and SP layouts with numbered items (1)–(16)
Validation + events	V-01 ... and E-01 to E-06: load, blur, input, save, results
Non-functional	WCAG 2.2 AA details, observability, performance



Video 01:16. BD-001 streams to the terminal.

Durable state lives in the work item, not the chat

CREATED

`plan.md` — revision 1, six milestones, marked **awaiting review**.

`status.md` — what is done, what is next, what is at risk.

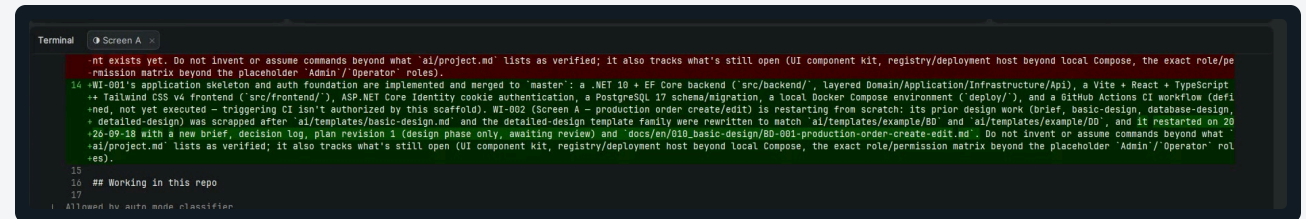
`evidence.md` — a traceability matrix from each REQ to design, code, test and status.

UPDATED SO PEOPLE CAN FIND IT

`docs/en/000_requirements/README.md` points to the brief.

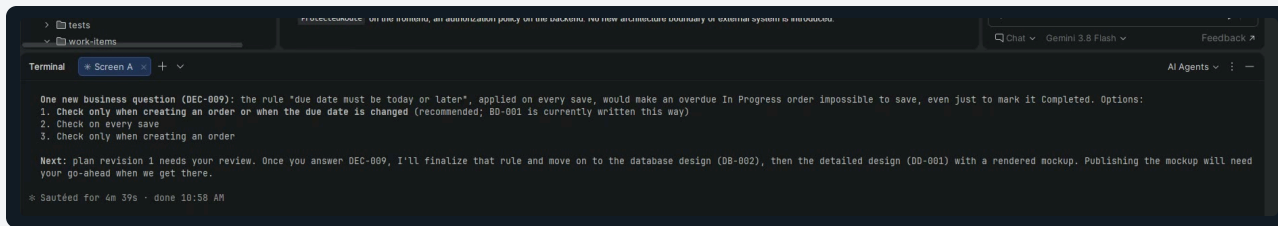
`docs/en/010_basic-design/README.md` lists BD-001.

`CLAUDE.md` "Current state" describes WI-002's progress.



Video 01:33. Traceability matrix and README updates.

Claude reports back and asks one question



Video 01:52. The end of the turn 1 summary — 4 m 39 s of agent work.

DEC-009 — ASKED IN THE SUMMARY

If "due date must be today or later" is checked on every save, an overdue In Progress order can't be saved, even just to mark it Completed.

1. Check only on create or when the due date changes (*recommended*)
2. Check on every save
3. Check only on create

OUTPUT

Summary + DEC-009 + "plan revision 1 needs your review". Next: database design (DB-002), then detailed design (DD-001) with a rendered mockup.

Claude Code pre-filled a suggested reply ("Go with option 1..."). **The user did not take it.**

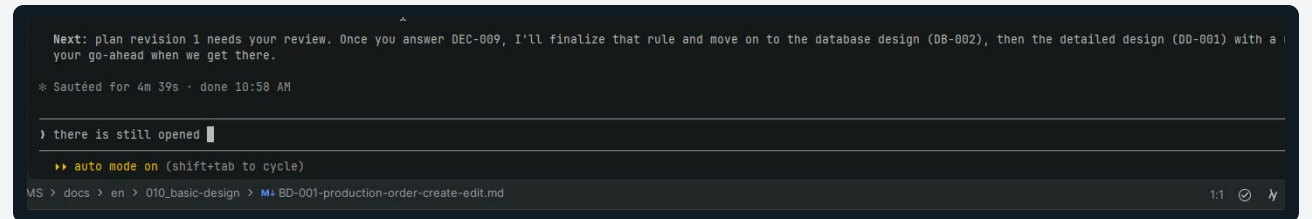
The human spots a gap

THE PROMPT

There are still open decisions to be made, right?

Claude had asked about only one open item. BD-001's own *Open questions* table listed five:

- Due-date rule for overdue orders (DEC-009)
- Two users editing the same order at once
- Which timezone defines "today"
- Order-number format
- CSRF protection for the new endpoints

A screenshot of a code editor with a dark theme. The editor shows a prompt in a light blue box: "Next: plan revision 1 needs your review. Once you answer DEC-009, I'll finalize that rule and move on to the database design (DB-002), then the detailed design (DD-001) with a your go-ahead when we get there." Below the prompt is a response in a light green box: "Sautéed for 4m 39s · done 10:58 AM". The response is followed by a line of code: "there is still opened". Below the code is a line of text: "auto mode on (shift+tab to cycle)". The editor's status bar at the bottom shows the file path: "MS > docs > en > 010_basic-design > BD-001-production-order-create-edit.md". The status bar also shows a zoom level of "1:1" and a search icon.

Video 02:09. The prompt is sent and Claude starts preparing the form.

Four questions, one form

Claude batched the open business questions into a tabbed multiple-choice form; each offered a recommended option.

DUE DATEDEC-009

When is "due date ≥ today" checked?
On create or when changed · every save · create only

CONCURRENCYDEC-010

Two users save the same order. Then what?
Reject the stale save · last write wins

TODAYDEC-011

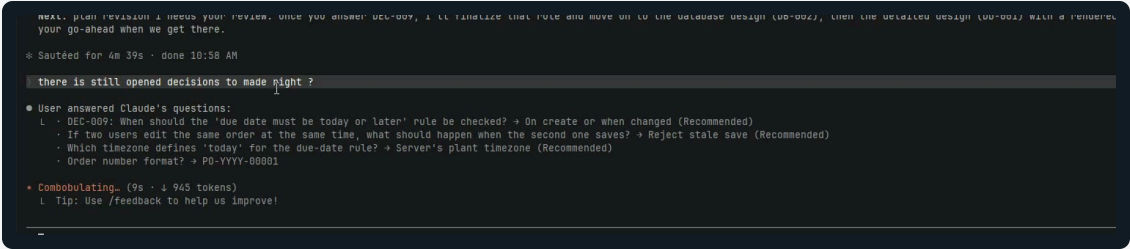
Which timezone defines "today"?
Server's plant timezone · the browser's

ORDER NO.DEC-012

Order number format?
PO-000001 · **PO-YYYY-00001** · plain number



Video 02:20. Tab 1: the due-date rule, with its consequence spelled out.



Video 02:42. Tab 4: the one where the user overrode the recommendation.

Every answer lands in the files, not just the chat

ID	DECISION	RECORDED OUTCOME
DEC-009	Overdue orders	Check only on create or when the due date changes
DEC-010	Concurrent edits	Reject a save based on stale data; the user reloads
DEC-011	"Today"	One configured plant timezone, applied server-side
DEC-012	Order number	PO-YYYY-NNNNN , sequence restarts each year

OUTPUT

`decisions.md` : DEC-009 to DEC-012 now decided. `brief.md` : REQ-010, REQ-011 and REQ-014 updated. **BD-001 v2**: new rule V-08 (stale-save check) and the new order-number format in the mockups. `status.md` : status **Green**; `plan.md` and `evidence.md` updated to match.

SELF-RECOVERY

A shell heredoc broke on apostrophes. Claude saved the script to a file and ran it from there, **with no help from the user.**

Business decisions now; technical ones where they belong

Claude closes the turn by separating out the technical choices — the ones that don't change how the screen behaves — and parks each at the design step that owns it:

DATABASE DESIGN — DB-002

How to generate a number sequence that restarts each year · what happens if a year passes 99,999 orders · how the system detects that someone else changed the order

DETAILED DESIGN — DD-001

Whether the new save endpoints need cross-site request forgery (CSRF) protection

"Plan revision 1 still needs your review. Once you approve it, I'll start DB-002."

```
I updated decisions.md, brief.md (requirements REQ-010, REQ-011 and REQ-014), and BD-001, which is now version 2 with V-08 added and the mockup numbers switched to the new format. status.md, evidence.md and plan.md are updated too, and the status is now Green.

A few technical choices remain for the next design steps. They don't change how the screen behaves:
- Database design (DB-002):
  - How to generate the number sequence that restarts each year.
  - What happens if a year goes past 99,999 orders.
  - How the system detects that someone else changed the order.
- Detailed design (DD-001): whether the new save endpoints need protection against cross-site request forgery (CSRF). I'll settle that while writing DD-001.

Plan revision 1 still needs your review. Once you approve it, I'll start DB-002.

* Worked for 1m 37s · done 11:02 AM
```

Video 03:25. The turn 2 summary — 1 m 37 s of agent work.

The plan waits for a human

#	MILESTONE	SKILL	DELIVERABLE
1	Brief + decision log	requirements	brief.md, decisions.md
2	Basic design	basic-design, screen-design	BD-001
3	Resolve DEC-009	—	decisions.md, BD-001 V-04
4	Database design	database-design	DB-002 schema doc
5	Detailed design + mockup	detailed-design, screen-design	DD-001 + rendered mockup
6	Reconcile and close	—	status.md, evidence.md

WHERE THE SESSION ENDS

Milestones 1–3 are done. Milestones 4–6 wait on the plan-approval gate in `ai/policies.md` : plans are reviewed before implementation continues.

FN-002	Load production order	Save changes to an existing order's evidence items	REQ-011, REQ-010
FN-003	Load production order	Fetch an existing order for display in edit mode	REQ-011
FN-004	List products	Provide the product choices for the product dropdown	REQ-015
FN-005	Validate order input	Enforce product, quantity, due date and notes rules (client for feedback, server authoritative)	REQ-013, REQ-014, REQ-015, REQ-016
FN-006	Enforce status transitions	Allow only the defined state-machine edges	REQ-017
FN-007	Enforce field lock by status	Reject product/quantity changes once the order is not <code>Draft</code>	REQ-018
FN-008	Enforce authentication and role	Reject unauthenticated (401) and non-Admin/Operator (403) requests	REQ-012

Actors and business flow

Actors: Admin, Operator — both already signed in through WI-001's login screen (DEC-001).

Create (UC-001)

1. User opens the create route → system loads the product list (FN-004) and shows an empty form; status shows `Draft`, order number shows "Assigned on save".
2. User selects a product and enters quantity, due date and optional notes → system shows field errors as fields lose focus (FN-005).
3. User presses `Save` → system validates all fields client-side; if valid, sends the request; the server re-validates (FN-005, FN-008) and saves (FN-001).
4. On success → system navigates to the edit route of the new order and shows "Production order [order number] created."

Video 03:43. BD-001's item events, E-01 to E-07 — the screen, specified event by event.

Design phase, first half done

FILE	STATE AFTER THE SESSION
work-items/WI-002/brief.md	Revision 1, REQ-010 to REQ-018, updated with the answers
work-items/WI-002/decisions.md	DEC-001 to DEC-012 decided
BD-001-...-create-edit.md	Version 2: rule V-08 and the new order-number format
work-items/WI-002/plan.md	Revision 1, awaiting review
work-items/WI-002/status.md	Green; next action is the plan review
work-items/WI-002/evidence.md	Traceability matrix started

Sources: the session recording ([ScreenA_BD.mp4](#)), its transcript, and the committed files in [work-items/WI-002/](#) .

NEXT IN THE SERIES

- **Database design (DB-002)** — sequence generation, the yearly reset and concurrency detection become schema decisions.
- **Detailed design (DD-001)** — API, functions, screen processing and a rendered mockup.
- **Implementation** — code and tests at every level.

HARNESS IMPROVEMENTS

None from this step. The **detailed design** and **implementation** steps each produced one (RFC 0001 and RFC 0002).

Screen A: database design

Three one-line prompts and one four-question form, and Claude Code produced a 216-line schema design, eight recorded decisions, a new requirement and a revision of the basic design — for the production-order create/edit screen.

Presented alongside `ScreenA_DB_edited.mp4` (3 min 59 s). Step numbers on the slides match the captions in the video.
Recorded 2026-09-18 · Claude Code (Opus 5) in the JetBrains Rider terminal, auto mode on.

`0002-production-order-schema.md`

`decisions.md`

`brief.md`

`BD-001-...-create-edit.md`

`plan.md`

`status.md`

`evidence.md`

Eleven minutes of recording, four and a half of agent work

The human wrote three prompts and answered one form. Everything else was the agent working through the harness.

3

prompts typed

each one line long

4

questions answered

in a single form

4 m 19 s

agent working time

2 m 15 s + 13 s + 1 m 51 s

8

decisions recorded

DEC-013 to DEC-020

The human

- Approved plan revision 1 and asked for the database design.
- Asked whether any open questions were left — the pivot of the session.
- Chose an answer for each of four questions. One was free text, and on the database-login split they **overrode the recommendation**.

The agent, guided by the harness

- Read the database skill, rules and template — and the persistence code — before writing anything.
- Decided three schema-level technical points itself, each recorded with its options and rationale.
- Corrected its own draft twice, unprompted.
- Reported what the user's choice would cost: a plan revision touching WI-001's Compose setup.

12 steps, one video

Each step is a clip in the edited video. **Amber: the human acts.** **Teal: the agent works and produces output.**

STEP 0100:00 Prompt 1 Approve the plan, start DB design	STEP 0200:12 Reads first Skill, rules, template, then the code	STEP 0300:32 DB-002 written 216 lines of schema design	STEP 0400:46 Self-correction Rewrites its own operational note
STEP 0500:59 Plan · status · evidence Approval written into the work item	STEP 0601:15 Turn 1 summary DEC-013–DEC-015 taken, DEC-016 asked	STEP 0701:33 Prompt 2 "Are there any open questions left?"	STEP 0801:44 Five open items Three of them previously unrecorded
STEP 0902:02 Prompt 3 "Let's answer all of them"	STEP 1002:11 Decision form Four tabs, then review and submit	STEP 1102:50 Recording the answers DEC-016–DEC-020 and REQ-019	STEP 1203:32 Review the schema decisions.md and DB-002

Times are positions in `ScreenA_DB_edited.mp4`. Long stretches of agent work are jump-cut to their result; nothing is sped up.

One line approves the plan and starts the schema

THE PROMPT

Approved the plan;
let's move on to DB design

Claude Code had pre-filled a suggested reply — *"plan approved, let work on the database design next"*. The user cleared it and typed their own. The prompt names no file, no template and no schema.

OUTPUT

No artifact yet. The approval itself is the output: it unlocks milestone 4 of plan revision 1, which the previous session stopped at.

```
... and the screen as new screen.

A few technical choices remain for the next design steps. They don't change how the screen behaves:
- Database design (DB-002):
  - How to generate the number sequence that restarts each year.
  - What happens if a year goes past 99,999 orders.
  - How the system detects that someone else changed the order.
- Detailed design (DD-001): whether the new save endpoints need protection against cross-site request forgery (CSRF). The first pass concluded no extra token was needed; I'll c

Plan revision 1 still needs your review. Once you approve it, I'll start DB-002.

* Worked for 1m 37s · done 11:02 AM

} approved the plan, let move on to DB design
>> auto mode on (shift+tab to cycle)
```

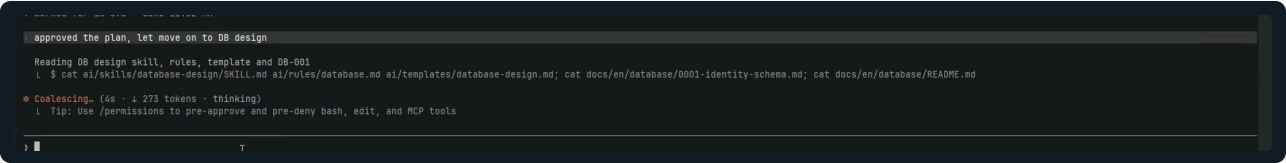
Video 00:06. The prompt replacing Claude Code's suggestion.

Read the rules, then read the code

WHAT CLAUDE READ	WHAT IT GOT FROM IT
database-design skill, ai/rules/database.md	The structure DB-002 must follow, and the bar: least privilege, justify every index, state migration impact
ai/templates/ database-design.md	The section order, including an <i>Open decisions</i> table
DB-001, database README	The conventions the identity schema already set
AppDbContext.cs, EF migrations	snake_case naming, uuid keys — what is actually built, not what the docs claim

OUTPUT

"Plan approved, so I'm recording that and starting DB-002. I've checked the existing persistence setup (snake_case naming convention, uuid keys, EF migrations) so DB-002 can follow it."



Video 00:18. Read-only commands, collapsed by Claude Code.

216 lines of schema, then Claude fixes its own sentence

STEP 3 — DOCS/EN/DATABASE/0002-PRODUCTION-ORDER-SCHEMA.MD

TABLE	PURPOSE
products	Minimal product reference data, seeded, no catalog UI
production_orders	One row per production order created on SCR-001
production_order_number_counters	Last issued sequence per year, to generate PO-YYYY-NNNNN

Plus constraints and indexes with a reason each, an API-field mapping for DD-001 to confirm, and a rollback command with the warning that rolling back after use deletes all order data.

```

192 -- **Rollback plan:** "Detest of database update InitialIdentitySchema --project src/backend/ProductionManagementAI.Infrastructure --startup-project src/backend/ProductionManagementAI.Api" drops the three tables
    nothing outside them references them.
193 -- **Operational note:** the generated 'order_number' column is written by raw SQL in the migration (EF Core 'HasComputedColumnSql(..., stored: true)'). The expression uses explicit '::text' casts because Postgr
    eSQL requires generation expressions to be immutable.
194 -- **Operational note:** the generated 'order_number' column is defined with EF Core 'HasComputedColumnSql(..., stored: true)'. The expression uses explicit '::text' casts because PostgreSQL requires generation
    expressions to be immutable.
195
196 -- Open decisions
197
```

Video 00:52. Step 4: Claude rewrites line 193 of its own draft.

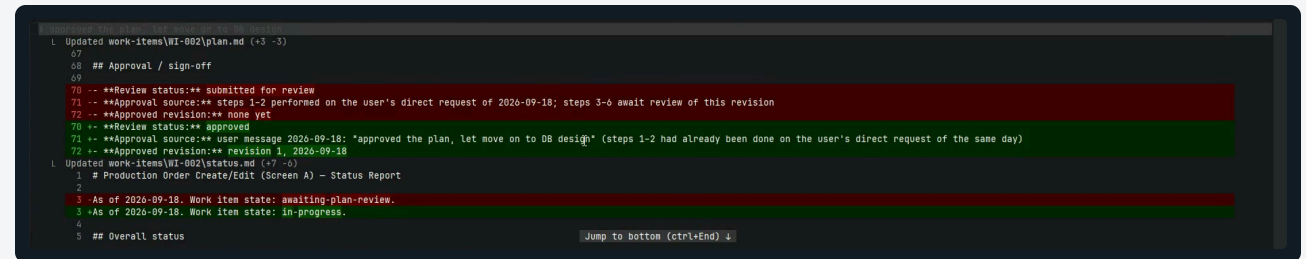
STEP 4 — SELF-CORRECTION

"...the generated order_number column is written by raw SQL in the migration" became "...is defined with EF Core HasComputedColumnSql(..., stored: true)". Nobody asked for it.

The approval is a file change, not a memory

- **plan.md** — "Review status: submitted for review" becomes **approved**; the "steps 3–6 await review" line is removed.
- **evidence.md** — the traceability matrix gains a DB-002 reference on every affected requirement row: `xmin` concurrency for REQ-011, the quantity CHECK for REQ-013, the product foreign key for REQ-015.
- **status.md** — the next milestone becomes DD-001; the risk "plan revision 1 not yet reviewed" is replaced by the new one, DEC-016.
- **decisions.md** — DEC-013, DEC-014 and DEC-015 recorded as decided, each with its options table and rationale.

Each edit is labelled "Allowed by auto mode classifier" on screen.



```
L Updated work-items\WI-002\plan.md (+3 -3)
67
68 ## Approval / sign-off
69
70 -- **Review status:** submitted for review
71 -- **Approval source:** steps 1-2 performed on the user's direct request of 2026-09-18; steps 3-6 await review of this revision
72 -- **Approved revision:** none yet
73 -- **Review status:** approved
74 -- **Approval source:** user message 2026-09-18: 'approved the plan, let move on to DB design' (steps 1-2 had already been done on the user's direct request of the same day)
75 -- **Approved revision:** revision 1, 2026-09-18
L Updated work-items\WI-002\status.md (+7 -0)
1 # Production Order Create/Edit (Screen A) - Status Report
2
3 As of 2026-09-18. Work item state: awaiting-plan-review.
3 As of 2026-09-18. Work item state: in-progress.
4
5 ## Overall status
Jump to bottom (ctrl+End) ↓
```

Video 01:07. Approval and traceability written into the work item.

Three decisions taken, one handed back

ID	DECIDED BY CLAUDE, BECAUSE IT DOES NOT CHANGE WHAT THE USER SEES
DEC-013	Order numbers come from a per-year counter table. A year's 100,000th order is rejected rather than changing the user's chosen format.
DEC-014	Use PostgreSQL's <code>xmin</code> to detect that someone else changed an order.
DEC-015	Status is stored as text with a CHECK on the four allowed values.

ONE OPEN ITEM — DEC-016

"Today the app and the migrations both connect to the database as the owner account. The database rules call for a separate app login with only the permissions it needs. Changing that means editing WI-001's Compose setup, so I recommend doing it before any deployment beyond your machine. **It doesn't block the design.**"

```
- A mapping from each column to a proposed API field, for DD-001 to confirm.
- Migration impact: it only adds tables. The doc includes the rollback command and notes that rolling back after use deletes all order data.

New decisions: I decided three technical points myself because they don't change how the screen behaves (details in decisions.md):
- DEC-013: order numbers come from the per-year counter table. A year's 100,000th order is rejected rather than changing your chosen format.
- DEC-014: use xmin to detect that someone else changed an order.
- DEC-015: status is stored as text with a check on the four allowed values.

One open item, DEC-016: today the app and the migrations both connect to the database as the owner account. The database rules call for a separate app login with only the permissions it needs. Changing that means editing
WI-001's Compose setup, so I recommend doing it before any deployment beyond your machine. It doesn't block the design.

status.md, evidence.md (traceability now includes DB-002), the database README, and DD-001's open-questions table are updated to match.

Next is step 5: the detailed design (DD-001) plus a rendered mockup. Publishing the mockup as a private artifact will need your go-ahead when I get there.

v Brewed for 2m 15s - done 11:00 AM
```

Video 01:24. The end of the turn 1 summary — 2 m 15 s of agent work.

The question that changed the session

THE PROMPT

Are there any open questions left?

Claude's summary had named one open item and implied that was the list. Instead of accepting it, the user asked the agent to go and check.

WHAT CLAUDE DID

Rather than answer from memory, it searched its own artifacts — `grep "\|open|proposed|unresolved"` across the work item and the design documents.

The same move as in the basic-design session, and it paid off the same way.

```
- DEC-014: use xmin to detect that someone else changed an order.  
- DEC-015: status is stored as text with a check on the four allowed values.  
  
One open item, DEC-016: today the app and the migrations both connect to the database as the owner account. The database rules call for a separate app login with only the perm:  
WI-001's Compose setup, so I recommend doing it before any deployment beyond your machine. It doesn't block the design.  
  
status.md, evidence.md (traceability now includes DB-002), the database README, and BD-001's open-questions table are updated to match.  
  
Next is step 5: the detailed design (DD-001) plus a rendered mockup. Publishing the mockup as a private artifact will need your go-ahead when I get there.  
  
* Brewed for 2m 15s · done 11:06 AM  
are there any open questions left ?  
* Doing.. (3s · thinking)  
> |
```

Video 01:38. The prompt is sent and Claude greps its own artifacts.

Two known — and three the agent had quietly decided

ITEM

- 1 DEC-016, splitting the database logins — already raised
- 2 CSRF protection — parked for DD-001 to settle
- 3 **Which plant timezone.** "You chose one configured plant timezone, but not which one."
- 4 **Cancel with unsaved changes.** BD-001 leaves the page without asking; should it confirm?
- 5 **Sample products.** Five placeholder products are seeded — swap them for something more realistic?

WHY IT MATTERS

Items 3 to 5 were gaps the agent had closed on its own without recording them as questions. Being asked to re-check surfaced them **before** the detailed design was built on top of them.

```
1. DEC-016, splitting the database logins. Today the app and the migrations both connect as the database owner. My recommendation is to keep that for the local demo and split them before any deployment beyond your machine. It needs your call eventually, but it doesn't block the detailed design.
2. CSRF protection (DD-001, open questions). This is a technical point that the detailed design (DD-001) is meant to settle. No extra token is needed (same-origin cookie with SameSite=Lax and no CORS). I'll confirm or revisit that while writing DD-001.

Rechecking just now, I also found three gaps that aren't recorded as open questions. I filled them in myself, and you should confirm them:
3. Which plant timezone. You chose one configured plant timezone (DEC-011), but not which one. It decides what "today" means for due dates and which year goes into PD-YYYY-... It needs a concrete value such as Asia/Ho_Chi_Minh or Asia/Tokyo.
4. Cancel with unsaved changes. BD-001 says Cancel just leaves the page without asking. A common alternative is to ask "Discard your changes?" when the form has been edited.
5. Sample products. DD-002 seeds five made-up products (P-1001 Steel bracket - P-1005 Hydraulic valve). They're placeholder demo data. Swap them if you want something more realistic for the videos.

Items 3 and 4 change what users see, so I'd like your answers before DD-001 builds on them. Tell me the timezone and whether Cancel should ask to confirm, and I'll record them and go on to DD-001.
```

Video 01:53. "I filled them in myself, and you should confirm them."

Four questions, one form

The user starts typing a direct answer to the two questions Claude asked for — "Asia/Ho_Chi_Minh, and yes, confirm on cancel" — then deletes it and asks for all five instead: **"Let's answer all of them"**.

TIMEZONEDEC-017

Which zone sets "today" and the year in PO-YYYY?

Ho_Chi_Minh · Tokyo · UTC

CANCELDEC-018

Cancel after editing the form — then what?

Confirm if edited · leave without asking

PRODUCTSDEC-019

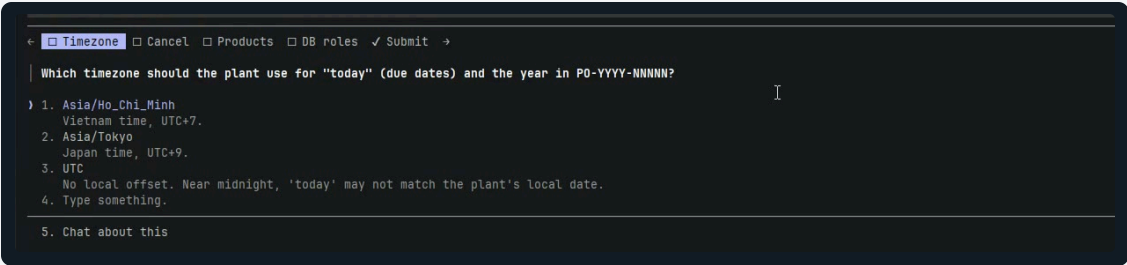
What should the demo seed as products?

5 · ~15 · free text: 30

DB ROLESDEC-016

How should database logins be handled?

split before non-local · split now



Video 02:16. Tab 1 of the form.



Video 02:35. A free-text answer: "Let's do 30 samples so we can get a sense of the screen's workload".

Every answer lands in the files, not just the chat

ID	QUESTION	RECORDED OUTCOME
DEC-016	Database logins	Split now: the owner login only runs migrations; the backend runs as a restricted <code>pmai_app</code> login, in local Compose too
DEC-017	Plant timezone	<code>Asia/Tokyo</code> , kept as a config value
DEC-018	Cancel with unsaved changes	Ask "Discard your changes?" only if the form was edited — becomes REQ-019
DEC-019	Demo product seed data	30 sample products, <code>P-1001</code> – <code>P-1030</code>
DEC-020	CSRF protection	Claude's own technical call: no extra token — <code>SameSite=Lax</code> cookie, no CORS, JSON-only bodies

CHECKED BEFORE WRITTEN

Before recording anything, Claude re-read the code the answers affect — the migration startup path, the cookie setup in `DependencyInjection.cs` and `deploy/compose.yaml` — then confirmed the cookie configuration matches what ADR-0002 assumed.

```

Sautted for 13s · done 11:07 AM
Let answers all of them
User answered Claude's questions:
  - Which timezone should the plant use for "today" (due dates) and the year in DD-YYYY-MMNNNN? → Asia/Tokyo
  - When the user presses Cancel after editing the form, what should happen? → Confirm if edited (Recommended)
  - What should the demo seed as products? → let do 30 sample so we could get a sense of the screen work load
  - DEC-016: How should database logins be handled? → Split now
Checking migration startup and cookie/CSRF setup
  - $ grep -rE "Migrate\|EnsureCreated\|IdentityServer" src/backend --include=*.cs | grep -v "/Migrations/" | head; sed -n 1,40p deploy/compose.yaml; ls deploy/docker
Brewing... (22s · ± 1.7k tokens · Thinking)
```

Video 02:56. The four answers, as Claude Code recorded them.

The choice is accepted — and its cost is reported

PLAN SCOPE

"Because the login split is happening now, the implementation plan (revision 2) will need to change WI-001's Docker Compose files: a new required runtime password in `.env.example` with no default; a second connection string; a script that grants the runtime login's permissions; and the migration command in `ai/project.md` updated to run as the owner login."

"That plan revision will come to you for review before any of it is built."

Updated files: `decisions.md`, `brief.md` (REQ-019 added), BD-001 (now revision 3), DB-002, ADR-0002, DB-001, `plan.md`, `status.md` and `evidence.md`.

```
later* note. You can override it.

Updated files: decisions.md, brief.md (REQ-019 added), BD-001 (now revision 3), DB-002, ADR-0002, DB-001, plan.md, status.md and evidence.md.

Plan scope: because the login split is happening now, the implementation plan (revision 2) will need to change WI-001's Docker Compose files:
- a new required runtime password in .env.example, with no default;
- a second connection string;
- a script that grants the runtime login's permissions;
- updating the migration command in ai/project.md to run as the owner login.

That plan revision will come to you for review before any of it is built.

Next is step 5: the detailed design (DD-001) plus a rendered mockup. I'll ask before publishing the mockup.

» Sautted for 1m 51s - done 11:12 AM
```

Video 03:23. "Nothing in Screen A's design is open now."

OUTPUT

Every decision from DEC-001 to DEC-020 is decided. The design phase's remaining work is DD-001 and a rendered mockup — and Claude states it will ask before publishing that mockup.

The open-decisions table, all resolved

The user reads `decisions.md`, then opens DB-002 and scrolls to the table the document opened with — the one that listed what was still undecided when the schema was written.

DECISION	STATUS
Per-year order sequence	decided — DEC-013
Past 99,999 orders in one year	decided — DEC-013
Optimistic-concurrency token	decided — DEC-014
Status storage	decided — DEC-015
Split migration-owner and runtime roles	decided — split now (DEC-016, user)
Demo product seed size	decided — 30 (DEC-019, user)

The next prompt, "Move on to the detailed design", is typed but not sent. The recording ends there.

src/backend/ProductionManagementAI.Api drops the three tables. Nothing outside them references them.

- Operational note: the generated `order_number` column is defined with EF Core `HasComputedColumnSql(..., stored: true)`. The expression uses explicit `::text` casts because PostgreSQL requires generation expressions to be immutable.

Open decisions

Decision	Options	Recommendation	Status
How to generate the per-year order sequence	per-year counter table with upsert; <code>MAX(order_seq)+1</code> with retry; one PostgreSQL sequence per year	Counter table	decided — see <code>work-items/WI-002/decisions.md</code> DEC-013
What happens past 99,999 orders in one year	reject the create; widen to 6 digits	Reject (CHECK constraint)	decided — DEC-013
Optimistic-concurrency token	PostgreSQL <code>xmin</code> ; explicit <code>version</code> column	<code>xmin</code>	decided — DEC-014
Status storage	<code>varchar</code> + CHECK; PostgreSQL enum type; smallint code	<code>varchar</code> + CHECK	decided — DEC-015
Split migration-owner and restricted runtime DB roles	split now; keep the single owner role for the local demo	Split before any shared/non-local deployment	decided — split now (DEC-016, user)
Demo product seed size	5; ~15; 30	5	decided — 30 (DEC-019, user)

Video 03:52. DB-002's open decisions, every row now carrying a decision ID.

Design phase, second half unlocked

FILE	STATE AFTER THE SESSION
0002-production-order-schema.md	Created; tables, constraints, grants, API mapping
decisions.md	DEC-001 to DEC-020 decided
brief.md	REQ-019 added
BD-001-...-create-edit.md	Revision 3
plan.md	Revision 1 approved; revision 2 scope known
status.md	Green; next action is DD-001
evidence.md	Traceability now reaches the schema

Sources: the session recording (`ScreenA_DB.mp4`), its transcript, and the committed files in `work-items/WI-002/` and `docs/en/database/` .

NEXT IN THE SERIES

- **Detailed design (DD-001)** — API contract, function and screen-processing designs, and a rendered mockup. Publishing the mockup needs the user's go-ahead.
- **Implementation** — including the restricted runtime login this session decided on, and code and tests at every level.

CARRIED FORWARD

DEC-016 turns into real work: a new password in `.env.example` , a second connection string, a grant script, and a migration command that runs as the owner.

Screen A: detailed design

Two prompts and one authorization question, and Claude Code produced four detailed-design documents, a rendered mockup published as a private artifact, four decisions — and a change to the harness that produced them.

Presented alongside `ScreenA_DD_edited.mp4` (4 min 00 s). Step numbers on the slides match the captions in the video.
Recorded 2026-09-18 · Claude Code (Opus 5) in the JetBrains Rider terminal, auto mode on.

`DD-001-production-order-create-edit.md`

`DD-001-API-production-orders.md`

`DD-001-FN-production-order-service.md`

`DD-001-SPD-...-create-edit.md`

`DD-001-screen-a-mockup.html`

`decisions.md · status.md · evidence.md`

Sixteen minutes of recording, ten and a half of agent work

The human wrote two prompts and answered one question. Everything else was the agent working through the harness.

2

prompts typed

one of them a correction

1

authorization asked

answered in 4 seconds

10 m 23 s

agent working time

7 m 27 s + 2 m 56 s

1,314

lines of design

across four documents

The human

- Reviewed the database design, then asked for the detailed design.
- Cleared the one permission the plan had deferred, before any work depended on it.
- Noticed that two of the four DD templates had produced no document — and said so in one line.

The agent, guided by the harness

- Read the plan's authorization table and stopped to ask, rather than discovering the gate halfway through.
- Read the running app's own CSS before designing a mockup for it.
- Fed a gap found while writing the API contract back up into the basic design.
- Turned its own mistake into a memory, then into a proposal to change the shared templates.

12 steps, one video

Each step is a clip in the edited video. **Amber: the human acts.** **Teal: the agent works and produces output.**

STEP 0100:00 Prompt 1 DB design reviewed, start the DD	STEP 0200:12 Reads first Skills, templates, and the app's own code	STEP 0300:32 Authorization May the mockup be published?	STEP 0400:50 DD-001 written The detailed design
STEP 0501:03 DD-001-API written Split out for Screen B to reuse	STEP 0601:16 Mockup approach Reuses the app's own palette	STEP 0701:38 Mockup published Private artifact, 11 states	STEP 0801:51 Turn 1 summary DEC-021–DEC-024, BD-001 rev 4
STEP 0902:23 Review: DD-001 ...and the DD template folder	STEP 1002:37 Prompt 2 Two templates produced no document	STEP 1102:49 Memory, then the fix DD-001-FN, DD-001-SPD, rewiring	STEP 1203:47 Review the four docs ...plus the mockup

Times are positions in `ScreenA_DD_edited.mp4`. Long stretches of agent work are jump-cut to their result; nothing is sped up.

Review first, then hand the work on

THE PROMPT

I reviewed the DB design and it looks okay, so let's move on to detailed design for this work item

Before typing, the user scrolls `plan.md`: its **Authorization** table ("Publish rendered mockup as a private Artifact — ask at execution — not yet authorized — step 5"), its risks, and the sign-off recording revision 1 as approved.

OUTPUT

The prompt carries a review result, not just an instruction. Claude Code's suggested reply ("Move on to the detailed design") was cleared and replaced.

```
plan scope: because the login split is happening now, the implementation plan (revision 2) will need to change WI-001's Docker compose files:
- a new required runtime password in .env.example, with no default;
- a second connection string;
- a script that grants the runtime login's permissions;
- updating the migration command in ai/project.md to run as the owner login.

That plan revision will come to you for review before any of it is built.

Next is step 5: the detailed design (DD-001) plus a rendered mockup. I'll ask before publishing the mockup.

* Sautéed for 1m 51s · done 11:12 AM

} i reviewed the DB design and it look okay so let move on to detailed design
  >> auto mode on (shift+tab to cycle)
```

Video 00:06. `plan.md` above, the prompt below.

Read the rules — and the running app

WHAT CLAUDE READ	WHAT IT GOT FROM IT
detailed-design and screen-design skills	The route through this step and its checklists
ai/templates/DD/*	The four DD templates and what each is for — the detail that matters in step 10
LoginPage.tsx, AuthContext.tsx, index.css	The app's real visual language — used an hour later to decide how the mockup should look
ADR-0001, the test projects	The layered backend contract and what test tooling already exists

OUTPUT

Claude does not start writing. It stops on a permission question instead — "Before writing: plan step 5 says publishing the rendered mockup as a private artifact needs your OK at that point, so I'm asking now so it doesn't hold things up later."



Video 00:18. Read-only commands, collapsed by Claude Code.

The gate was written down, so it was honoured

```
Next is step 5: the detailed design (DD-001) plus a rendered mockup. I'll ask before publishing the mockup.

* Sautéed for 1m 51s · done 11:12 AM

i reviewed the DB design and it look okay so let move on to detailed design for this work item

Read 3 files, listed 1 directory, ran 3 shell commands

• Before writing: plan step 5 says publishing the rendered mockup as a private artifact needs your OK at that point, so I'm asking now so it doesn't hold things up later.

· Photosynthesizing... (1m 12s · ↓ 6.6k tokens)

> |

>> auto mode on (shift+tab to cycle) · esc to interrupt · ← for agents
```

Video 00:41. One question, two real options, answered in about four seconds.

WHAT WAS ASKED

"DD-001 includes a rendered mockup of Screen A covering create, edit, locked-fields, validation-error, success and discard-dialog states. Can I publish it as a private claude.ai artifact (only you can see it unless you share it)?" — **Yes, publish privately** · Local HTML file only · Type something

WHY IT MATTERS

Nobody re-stated the rule in the prompt. It was a row in `plan.md`'s authorization table — *ask at execution* — written when the plan was drafted. The agent read it, honoured it, and cleared it **before** starting the work that depended on it, not halfway through.

The design, and the contract split out from it

STEP 4 — DD-001-PRODUCTION-ORDER-CREATE-EDIT.MD

609 lines: modules, screen states, processing flow, persistence, exception handling, observability and test viewpoints. Its tables map every BD rule to a concrete place in the code — `ProductionOrder.ValidateQuantity` for V-02/V-05, `IPlantClock.Today` for V-04, request-vs-entity `xmin` for V-08.

```
1
2
3 # Production Order Create/Edit - Detailed Design Document (詳細設計書)
4
5 DD-001 - Implements BD-001 (SCR-001), requirements REQ-010-REQ-019.
6
7 ## Document control (改訂履歴)
8
9 | Field | Value |
10 | --- | --- |
... +599 lines
```

Video 00:56. DD-001 streams to the terminal.

STEP 5 — DD-001-API-PRODUCTION-ORDERS.MD

292 lines. Split into its own document **because Screen B will reuse `/api/products`** — a decision about the next work item, made while writing this one.

```
1
2
3 # Production Orders API - API Specification Design (API仕様設計)
4
5 DD-001-API - supports DD-001, requirements REQ-010-REQ-018.
6
7 ## Document control (改訂履歴)
8
9 | Field | Value |
10 | --- | --- |
... +282 lines
```

Video 01:09. The API specification.

DESIGN FEEDS BACK UPSTREAM

Writing the contract exposed a missing quantity upper bound. It became **DEC-024** and **BD-001 revision 4** — not a silent choice in the code.

It looks like the product, because it read the product

STEP 6 — BEFORE BUILDING

"The DD documents are written. Before building the mockup I need to load the artifact design guidance." → `Skill(artifact-design)`

...AND THE VISUAL DECISION

"I'll build the mockup using the app's existing light-only Tailwind gray palette from `LoginPage.tsx` — gray-900 buttons, gray-300 borders, system font — while only the spec-sheet chrome around it adapts to the viewer's theme."

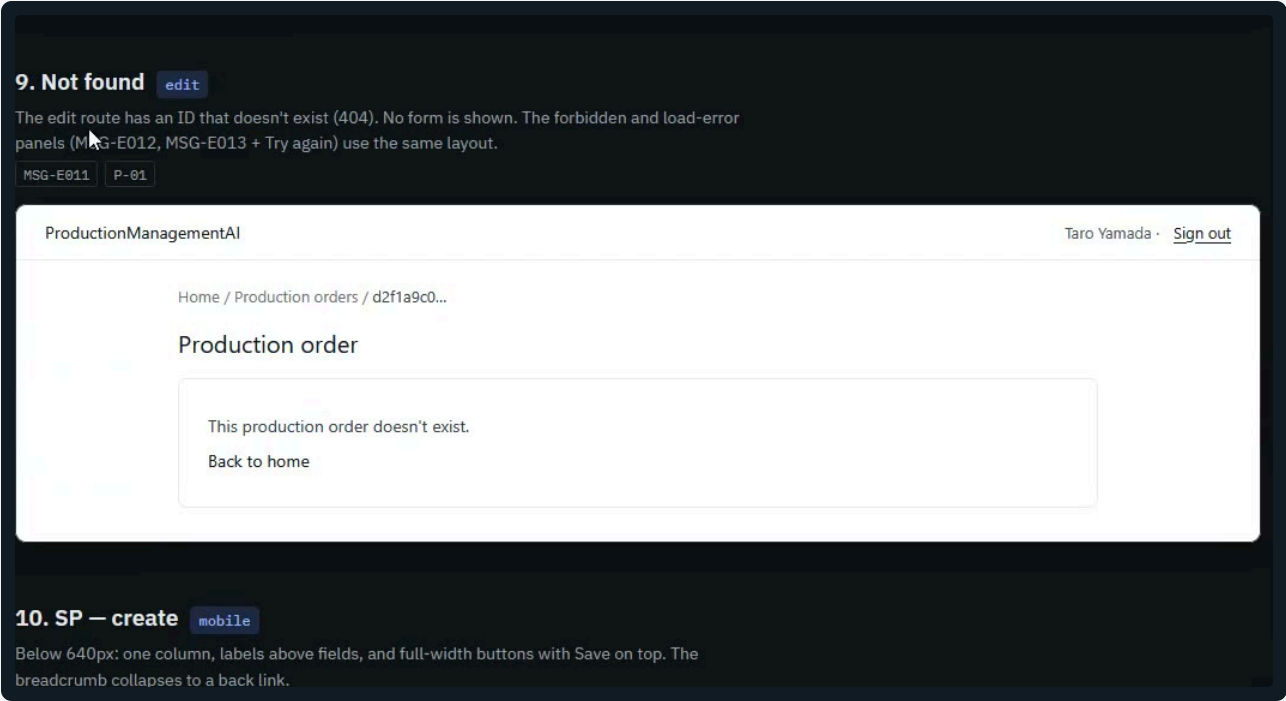
That is what reading `index.css` in step 2 was for.

```
• Publishing the mockup privately now.
• Artifact[D:\Work\AI\WMS\docs\en\020_detailed-design\mockups\DD-001-screen-a-mockup.html]
  ↳ Published 📄 https://claude.ai/artifact/FtG18d27Ujexv5C9jx0Mq
• Published. Next: linking it from DD-001, fixing the order of DD-001's revision table, and updating status and evidence.
  ↳ Photosynthesizing... (on 54s · ↓ 40.1k tokens)
✓ Update installed · Restart to update
```

Video 01:44. Step 7: published to a private artifact URL.

`mockups/DD-001-screen-a-mockup.html` (292 lines) stays in the repository; the artifact is the rendered copy, linked from DD-001.

A design a non-developer can actually judge



The artifact holds **11 static states**: create empty, product list open (30 items), validation errors, created → edit (Draft), in progress (locked), completed (terminal, overdue), stale save (409), discard dialog, not found, SP create, SP locked. Each carries the requirement and decision IDs it realises.

ID	DECIDED WHILE WRITING THE DESIGN
DEC-021	Browser-local date for early feedback; the server's plant date is authoritative
DEC-022	Native <code><select></code> ordered by SKU for 30 products
DEC-023	RFC 9457 error contract; <code>code</code> = message ID
DEC-024	Quantity upper bound 999,999,999 → BD-001 revision 4

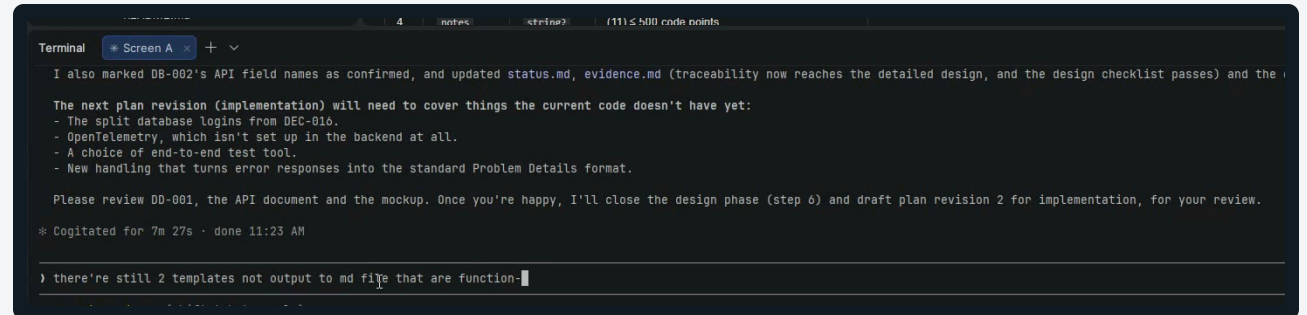
Video 01:58. State 8 of 11 — the discard dialog, tagged REQ-019, DEC-018, items 17–19, E-07, E-09.

The user almost accepted it

The user types a prompt accepting the work — "*The design looks okay, let's move on to the implementation*" — and **does not send it**. They read DD-001, then expand `ai/templates/DD/` in the project tree, where `api-design.md`, `function-design.md` and `screen-processing-design.md` sit side by side. Two of them produced no document.

THE PROMPT THAT WAS SENT INSTEAD

There are still 2 templates not output to an .md file: function-design and screen-processing-design



```
Terminal 4 notes: string? (11) 500 code points

I also marked DB-002's API field names as confirmed, and updated status.md, evidence.md (traceability now reaches the detailed design, and the design checklist passes) and the

The next plan revision (implementation) will need to cover things the current code doesn't have yet:
- The split database logins from DEC-010.
- OpenTelemetry, which isn't set up in the backend at all.
- A choice of end-to-end test tool.
- New handling that turns error responses into the standard Problem Details format.

Please review DD-001, the API document and the mockup. Once you're happy, I'll close the design phase (step 6) and draft plan revision 2 for implementation, for your review.

* Cogitated for 7m 27s · done 11:23 AM

} there're still 2 templates not output to md file that are function-
```

Video 02:43. The template folder open, and the correction being typed.

Fix the output, fix yourself, then propose the rule

"Understood. I left those two out as optional, but you want the full template family produced. **First I'll save that preference** so later design steps include all four."

SCOPE 1 — THIS AGENT

A memory, written first

Before producing anything, Claude wrote the preference down: *"in the WMS project, detailed design must output every DD template (main + api-design, function-design, screen-processing-design) as its own .md file"*. The order matters — the habit is fixed before the symptom.

SCOPE 2 — THIS WORK ITEM

The two missing documents

`DD-001-FN` (325 lines) and `DD-001-SPD` (212 lines). DD-001 was then rewritten to **point at them instead of repeating their content**, and the API document's cross-references, the docs index, `status.md` and `evidence.md` were updated to match.

SCOPE 3 — EVERYONE

A proposal, not a change

"The project's shared templates and the detailed-design skill still call the three companions optional. If you want that to be the rule for everyone, I can propose the change. **Changes to shared templates need your review before they take effect.**"

```
• Understood. I left those two out as optional, but you want the full template family produced. First I'll save that preference so later design steps include all four.

Searching for "###"
1 Remembering in the WMS project, detailed design must output every DD template (main + api-design, function-design, screen-processing-design) as its own .md file
2 "###"

• Bootstrapping... (10s - 1.0k tokens)
  ↳ Tip: Working with HTML/CSS? Install the frontend-design plugin:
    /plugin install frontend-design@claude-plugins-official

✓ Update installed · Restart to update
```

Video 02:56. "Searching for 1 pattern, writing 1 memory..."

```
1 DD-001-FN - used by DD-001 and DD-001-API (and later by Screen B's list DD), requirements REQ-010-REQ-010.
2
3 ## Document control (历史回顾)
4
5 | Field | Value |
6 | --- | --- |
7 - 324 lines

• Now the screen-processing document. I'll reread the current flows (P-01-P-05) in DD-001 so I can jump to bottom [ctrl+End] ↓ anything.
```

Video 03:22. The second companion, then DD-001 rewired to both.

The design set stops repeating itself

DOCUMENT	LINES	ROLE
DD-001	485	The design; now delegates to the three companions
DD-001-API	292	The HTTP contract; reusable by Screen B
DD-001-FN	325	Function design, backend and frontend
DD-001-SPD	212	Screen processing and exception handling

OUTPUT

"I saved this as a preference in memory: for this project, detailed design always produces all four template documents." — *Worked for 2m 56s*

```
- API document: its processing steps now point to the matching sections in the two new documents.
- docs/en/020_detailed-design/README.md, status.md and the traceability matrix in evidence.md now include the new files.

I saved this as a preference in memory: for this project, detailed design always produces all four template documents.

The project's shared templates and the detailed-design skill still call the three companions optional. If you want that to be the rule for everyone, I can propose the change to those files. Changes to shared templates need your review before they take effect.

Next, as before I once you've reviewed the four detailed-design documents and the mockup, I'll close the design phase and draft plan revision 2 for implementation.

* Worked for 2m 56s · done 11:28 AM
```

Video 03:38. The turn 2 summary, ending on the offer to change the shared templates.

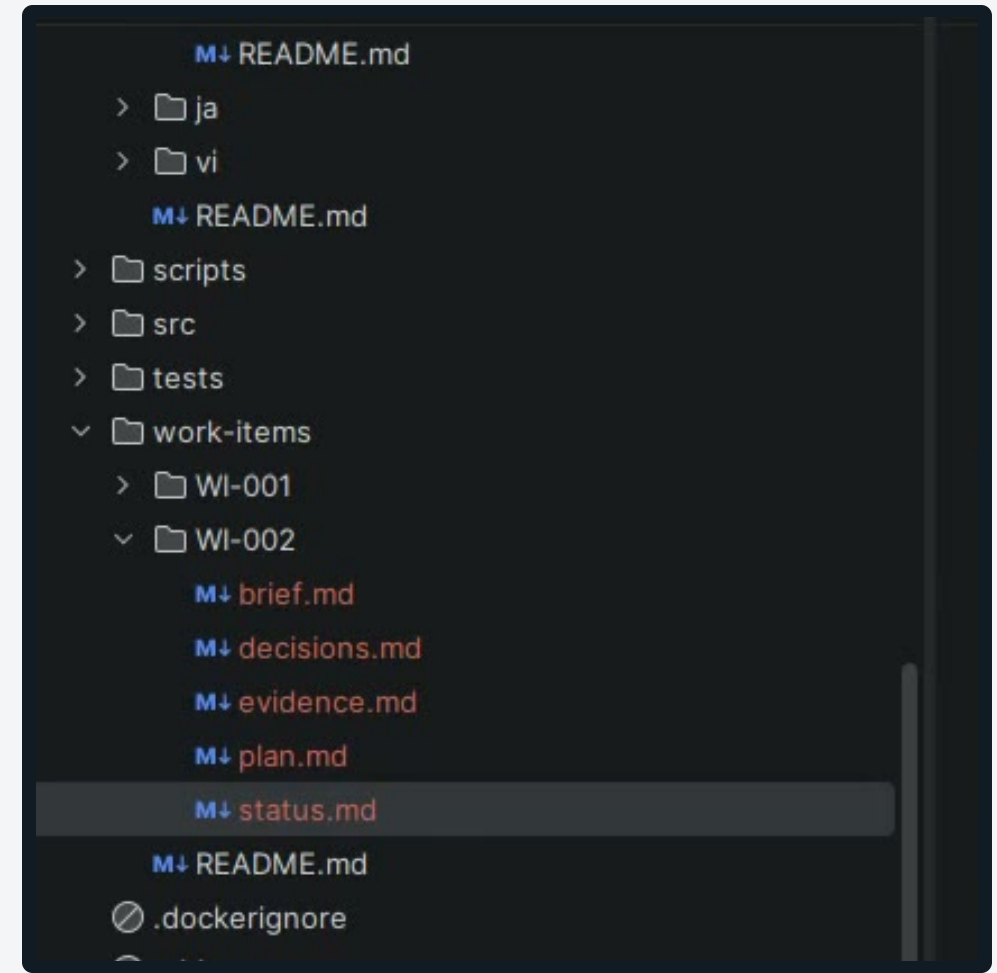
Four documents and a mockup, all traceable

The user reads each document in turn, ending on DD-001-SPD's exception-handling table — 401, 403, 404, 415, 400 with message IDs, 409 on a stale save, 422 on a rule violation, and 500 `MSG-E013` with no exception detail leaked.

Its closing section reads **"Unresolved decisions: None."** with the related decision IDs listed — DEC-007, DEC-008, DEC-009, DEC-018, DEC-020, DEC-021, DEC-023.

WHERE THE SESSION ENDS

Claude's suggested next prompt — "Update the templates and skill so all DD companions are always produced" — sits unsent in the input line. That became harness RFC 0001.



Video 03:53. 020_detailed-design, complete.

RFC 0001 — always produce all four DD documents

This is the one step in the Screen A series where using the harness exposed a defect **in the harness**, and the run produced a change proposal against it.

ROOT CAUSE — IN THE WORDING, NOT THE AGENT

The templates described the three companions as optional — "Use it when...", "None — this screen's design fits entirely in this file". Folding their content into the main DD was a *permitted reading* of the rule. The RFC records that, with the session as evidence.

WHAT THE RFC CHANGED

- `ai/templates/detailed-design.md` and the three `ai/templates/DD/*` templates
- `ai/templates/README.md`, `ai/skills/detailed-design/SKILL.md`
- `ai/evaluations/baseline-cases.md` — so the case is checked in future evaluations

THE BOUNDARY THAT WAS KEPT

The agent changed **its own** behaviour immediately, and **offered** the change to everyone's. Shared harness files are reviewed by their owner before they take effect — the same gate as the mockup in step 3, applied to the harness itself.

STATUS

Adopted. Recorded at `ai/improvements/0001-dd-companions-always-produced.md`.

One other walkthrough produced an RFC: the implementation step found that nothing told the harness to keep superseded plan revisions, which became **RFC 0002 — keep every plan revision, oldest first**.

Design phase complete, implementation next

FILE	STATE AFTER THE SESSION
DD-001 + API + FN + SPD	Complete; 1,314 lines, cross-linked
DD-001-screen-a-mockup.html	In the repo; published as a private artifact
decisions.md	DEC-001 to DEC-024 decided
BD-001-...-create-edit.md	Revision 4
status.md	Green; step 6 and the DD review remain
evidence.md	Traceability reaches the detailed design
ai/improvements/0001-...md	Adopted

Sources: the session recording ([ScreenA_DD.mp4](#)), its transcript, and the committed files in [docs/en/020_detailed-design/](#) , [work-items/WI-002/](#) and [ai/improvements/](#) .

CARRIED INTO THE IMPLEMENTATION PLAN

- The split database logins from DEC-016.
- OpenTelemetry, which isn't set up in the backend at all.
- A choice of end-to-end test tool.
- Turning error responses into the standard Problem Details format.

NEXT IN THE SERIES

Implementation — code and tests at every level, and the step that produced the second harness RFC.

Screen A: implementation

Six prompts and one form turned four design documents into a working screen: backend, database, frontend, 133 tests, two pull requests — and the harness improvements the earlier steps had only proposed.

Presented alongside `ScreenA_Implementaion_edited.mp4` (5 min 59 s, cut from 50 min 44 s). Step numbers on the slides match the captions in the video.

Recorded 2026-09-18 · Claude Code (Opus 5) in the JetBrains Rider terminal, auto mode on.

Domain · Application

Infrastructure · Migrations

Api · OpenTelemetry

frontend/production-orders

tests/e2e (Playwright)

plan.md · test-plan.md · evidence.md

ai/improvements/0001 · 0002

Fifty minutes of recording — and one 34-minute turn

The shape is unlike the other three steps. Three quarters of the agent's working time is a single unattended turn, so the human's work sits at both ends: the plan before it, the merge after it.

5

prompts in the cut

two of them corrections

34 m

longest single turn

domain → E2E, unattended

133

tests written

49 + 38 + 38 + 8

2

pull requests

Screen A and the harness

The human

- Chose the E2E tool, the accessibility approach, the git scope and where harness changes go.
- Made Claude restore the superseded plan — twice, until it was in chronological order.
- Approved revision 2, then squash-merged both PRs personally. **Merge was never authorized for the agent.**

The agent, guided by the harness

- Wrote RFC 0002 with its own evaluation case, and had already applied RFC 0001 before the cut begins.
- Named its own deviations: the index, the Docker workaround, an extension method that departed from the DD.
- Proved the restricted login instead of asserting it, and prevented a CRLF failure before it happened.
- Stopped at the merge gate and reconciled `master` afterwards.

11 steps, one video

Each step is one or more clips in the edited video. **Amber: the human acts.** **Teal: the agent works and produces output.**

STEP 0100:00 Four questions E2E · a11y · git scope · harness PR	STEP 0200:32 Plan revision 2 14 steps, and its own caveats	STEP 0301:16 Keep the old plan Restored — in the wrong place	STEP 0401:42 Chronological order The rule fixed, then the file
STEP 0502:08 Approved + RFC 0002 Keep every plan revision	STEP 0602:32 Branches, worktrees Two branches, PRs to come	STEP 0702:44 Backend + database Tests, CI verdict, CRLF caught	STEP 0803:29 API and frontend A real bug found and fixed
STEP 0903:58 E2E and the proof 8 journeys; PR opened; 34m turn ends	STEP 1005:17 The screen exists Both PRs squash-merged	STEP 1105:35 Reconciling master Two things need your call	NOT IN THE CUT The session also opened by applying RFC 0001 from the previous step — see slide 13

Times are positions in `ScreenA_Implementaion_edited.mp4`. 50 min 44 s of source, jump-cut to 5 min 59 s; nothing is sped up. The cut begins at the implementation planning — the session’s opening minutes, which applied the previous step’s RFC 0001, are left to that step’s deck.

The choices the agent would not make alone

E2E TOOL

DEC-025

Which end-to-end test tool?

Playwright · Cypress · none for now

A11Y CHECK

DEC-026

How should accessibility checks run?

axe in Vitest + E2E · manual only

GIT SCOPE

DEC-027

Which git operations are authorized?

branch, commit, push, open PR — **merge: no**

HARNESS PR

DEC-028

Where does the harness change go?

Its own branch/PR · same as Screen A

← ☒ E2E tool ☐ A11y check ☐ Git scope ☐ Harness PR ✓ Submit →

| Which end-to-end test tool should Screen A use? No E2E tool has been chosen yet; WI-001 deferred it.

1. Playwright (Recommended)
@playwright/test driving Chromium against the Docker Compose stack. Covers the create → edit and discard-dialog journeys; can also run automated accessibility (axe) checks.
2. Cypress
Also works well, but only runs JavaScript in the browser and has weaker multi-tab support.
- › 3. No E2E for now
Rely on unit + integration tests; record E2E as not run. The DD's E2E test scenarios stay uncovered.
4. Type something.

5. Chat about this

Enter to select · Tab/Arrow keys to navigate · Esc to cancel

Video 00:06. "No E2E tool has been chosen yet; WI-001 deferred it." — the question carries its own history.

A plan is a document, not a promise

14 numbered steps, each with what it depends on, which skill it uses, what it must deliver and **how that will be checked**. This is the artifact the rest of the session is measured against.

Deliverables and milestones

#	Milestone / step	Depends on	Skill used	Deliverable	Verification method
1	Harness branch: create worktree <code>../WMS-worktrees/harness-dd-companions</code> on <code>feature/harness-dd-companions</code> from <code>master</code> ; copy the RFC 0001 changes (<code>ai/templates/**</code> , <code>ai/skills/detailed-design/SKILL.md</code> , <code>ai/evaluations/baseline-cases.md</code> , <code>ai/improvements/**</code>); commit	none	harness-improvement	commit on the harness branch	<code>git diff master --stat</code> shows only <code>ai/</code> files
2	Screen A branch: create worktree <code>../WMS-worktrees/WI-002</code> on <code>feature/WI-002-production-order-screen</code> from <code>master</code> ; copy the WI-002 design files (work item, docs, <code>CLAUDE.md</code> , READMEs, ADR-0002/DB-	none	implementation	commit on the WI-002 branch	Diff contains only WI-002 design files

The plan itself

14 steps, each with a deliverable and a check

Video 00:55. Steps 1 and 2: the two branches — created before a line of Screen A is written.

THINGS TO KNOW BEFORE APPROVING

CI — opening the PRs starts GitHub Actions; the project's first real run. **E2E stays out of CI** — it needs the whole Compose stack. **Five new dependencies**, pinned and vulnerability-checked. **Branch name** — the skill suggests one, but the project's own DEC-009 wins; that contradiction was later fixed in RFC 0003.

(name)

Roles and responsibilities

Role	Owner
Plan author, implementer	this agent (Claude)
Reviewer, approver, merge decision	ThanhTN

Resources and external actions

Video 01:09. Who owns what, written down before the work starts.

Two corrections, one minute of typing

PROMPT 1

Keep the design plan in there so we can back-track it

RESULT

Revision 1 **rebuilt from the session's own edits** — git could not help, it had never been committed — and restored in full, with a note saying so. But placed **below** revision 2.

PROMPT 2

No — put it in chronological order

RESULT

"Got it: oldest first, so the file reads in the order things happened. I'll **update the saved rule first, then reorder the file.**"

```
- Revision 1: design phase, complete and superseded. It keeps all of its sections, including the Outcome column and the closing note. It still carries the note that I rebuilt it from this session because it was never committed.
- Revision 2: implementation, current. Its content is unchanged. Its opening now says revision 1 is "kept in full above".

Each revision is a top-level section in the file, with its usual sections one level below. Future revisions will be added at the bottom.

I updated the saved rule to match: every revision is kept in full, oldest first, and a new revision goes after the last one rather than above it.

Revision 2 still needs your approval. Once you give it, I'll start with step 1.

* Brewed for 23s · done 11:41 AM
```

✓ Update installed · Restart to update

Video 02:01. "Every revision is kept in full, oldest first, and a new revision goes after the last one."

Approval and a rule change, in one prompt

PROMPT 3

OK, the revision 2 plan seems solid, so approved. Also update the harness

RFC 0002 is written the same way as RFC 0001 — status, affected files, summary, and a motivation section citing this session as evidence.

THEN	WHAT WAS CREATED
Worktree 1	<code>feature/harness-wi002-feedback</code> — RFC 0001 and RFC 0002, one commit each
Worktree 2	<code>feature/WI-002-production-order-screen</code> — the design-phase commit
Main worktree	Cleaned; each copied file verified with <code>git diff --no-index</code> before reverting

```
3 **Status:** adopted
4 **Affected:** 'ai/templates/plan.md', 'ai/skills/planning/SKILL.md', 'ai/evaluations/baseline-cases.md'
5
6
7
8 'plan.md' keeps every revision of a work item's plan, in full and in chronological order. A new revision is appended after the last one. The previous revision is closed in place (Outcome column, Closure Line), and a revision index table at the top marks the current one.
9
10 ## Motivation
11 ~ +00 Lines
12
13 * Contemplating... (in 1s · ↓ 5.9k tokens)
```

Video 02:26. RFC 0002 — while the editor shows the plan's authorization table.

Diagnosing CI, and preventing a failure

```
• Step 4 is done: 35/35 unit tests pass, including the 4 existing ones. A quick check on the harness PR's CI run:  
  Ran 2 shell commands  
• Both CI jobs failed after 1 second. That points to setup, not code: the harness PR only changes Markdown. Checking the logs.  
  Ran 2 shell commands  
• CI can't run since GitHub rejected both jobs due to a billing lock on the account — that needs to be cleared on your end. I'll note this as blocked in evidence.md and move on to step 5: adding the two packages needed at the application layer and checking them for known vulnerabilities.  
  Ran 2 shell commands
```

Video 02:52. "That points to setup, not code: the harness PR only changes Markdown."

DIAGNOSED, NOT RETRIED

Both CI jobs failed in one second. Claude reasoned from the content — a Markdown-only PR cannot fail on code — read the logs, and found a **GitHub billing lock**. Recorded as blocked in `evidence.md` rather than treated as a defect.

```
• The solution builds with 0 warnings. Recording the index deviation in DB-002 and decisions.md now, while it's fresh.  
  Ran 2 shell commands  
• Moving to step 7: splitting the database logins. First I'll check the current Compose, env file, and git line-ending setup, since CRLF endings in shell scripts can break inside the Postgres container.  
  Ran 2 shell commands  
• There's no .gitattributes file, and with core.autocrlf set to true, the new init script would get CRLF line endings and fail on Linux. I'll add a .gitattributes rule to keep *.sh files as LF alongside the script.  
  Ran 2 shell commands  
• Now Compose: the database container gets the init script and the new required password setting, and the backend connects as psql_app instead of the owner.
```

Video 03:20. A Windows-to-Linux failure caught before it could happen.

PREVENTED, NOT DEBUGGED

"There's no `.gitattributes` file, and with `core.autocrlf` set to true, the new init script would get CRLF line endings and **fail on Linux**. I'll add a `.gitattributes` rule to keep `*.sh` files as LF alongside the script."

The requirement wrote the test; the test found the bug

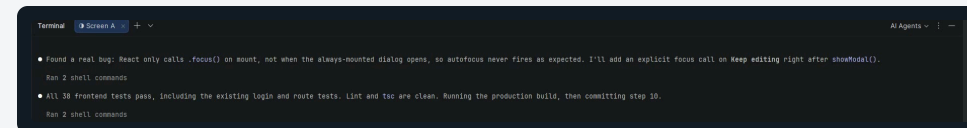
THE COMPONENT CITES ITS REQUIREMENT

// BD-001 items 17-19 (REQ-019, DEC-018). The native <dialog> + showModal() gives the focus trap, Escape and backdrop for free; Escape fires "cancel", which is treated as Keep editing.

A REAL BUG

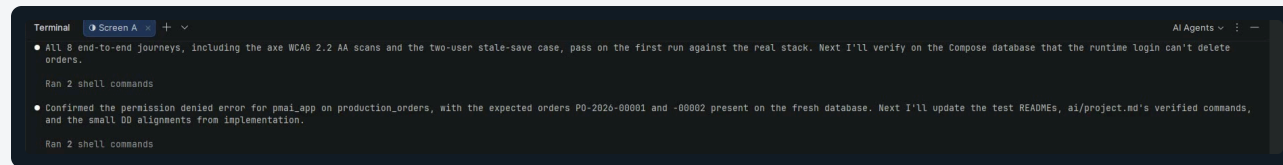
"React only calls `.focus()` on mount, **not when the always-mounted dialog opens**, so `autofocus` never fires as expected. I'll add an explicit focus call on **Keep editing** right after `showModal()`."

REQ-019 said focus must start on *Keep editing*. The component looked correct and was wrong. The only reason it was caught is that a test was written from the requirement.



Video 03:49. Found, explained and fixed — then 38 frontend tests pass.

The least-privilege decision, proved



```
Terminal Screen A + - AI Agents
• All 8 end-to-end journeys, including the axe WCAG 2.2 AA scans and the two-user stale-save case, pass on the first run against the real stack. Next I'll verify on the Compose database that the runtime login can't delete orders.
  Ran 2 shell commands
• Confirmed the permission denied error for pmai_app on production_orders, with the expected orders PO-2026-00001 and -00002 present on the fresh database. Next I'll update the test READMEs, ai/project.md's verified commands, and the small OD alignments from implementation.
  Ran 2 shell commands
```

Video 04:29. DEC-016, closed by attempting the forbidden operation.

"All **8 end-to-end journeys**, including the axe WCAG 2.2 AA scans and the two-user stale-save case, **pass on the first run** against the real stack."

"Confirmed the **permission denied** error for **pmai_app** on **production_orders**, with the expected orders **PO-2026-00001** and **-00002** present on the fresh database."

CAREFUL WITH SOMEONE ELSE'S MACHINE

The database volume was wiped **only because step 7 authorized it**. A fresh **PMAI_APP_DB_PASSWORD** was generated into the worktree's own **.env**, never printed, and checked as gitignored — the user's own **.env** was left untouched.

What an unattended half-hour reports back

KIND	WHAT THE SUMMARY SAID
Deviation	EF Core always adds an index on <code>product_id</code> ; removing it would make migrations drift, so DB-002 was updated instead (DEC-029)
Workaround	Docker Desktop won't mount files from the worktree, so the login script is baked into a small Postgres image — "also more portable"
Defect	The discard dialog wasn't focusing "Keep editing". Fixed and covered by tests
Noise	A Kerberos-library warning from the database driver; harmless, recorded in <code>evidence.md</code>
Limits	E2E isn't in CI yet and runs on Chromium only. No manual screen-reader pass

13	Full verification run from a clean environment; security-review and delivery checklists; <code>evidence.md</code> , <code>status.md</code>	12	security-review, pr-review	evidence, status	Checklists recorded pass/fail with reasons	done 2026-09-18 — final run 03.12Z all green; sr review and delivery checklists pass (<code>evidence.#</code>
14	Push both branches and open two PRs to master. (Screen A; harness). Watch the triggered CI runs and record the results. Merge is not done	13 (Screen A); 1 (harness)	pr-review, ci-cd	2 PRs	PR links + CI results in evidence	done 2026-09-18 — PR #3 (Screen A) https://github.com/thantrn95/ProductionManager and PR #2 (harness) opened; CI jobs not started (GitHub billing lock); GitHub secret scan par both; not merged (not authorized)

Done and responsible

Video 05:03. The same plan, hours later: every step now carries its Outcome.

```
Terminal Screen A
- An extra index (DEC-029): EF Core always adds an index on product_id, which DB-002 had left out. Removing it would make the migrations drift, so I kept it and updated DB-002.
- Database init script: Docker Desktop won't mount files from the worktree folder, so the script that creates the restricted login is built into a small Postgres image instead. It's also more portable.
- A real bug the tests caught: the discard dialog wasn't putting focus on "Keep editing" when it opened. It's fixed and covered by tests.
- A harmless log line: the backend container logs a Kerberos-library warning from the database driver. The app works normally; it's recorded in evidence.md.

What you need to do locally:
- Your old database is gone. Your local database volume was wiped, as the plan authorized, and recreated for the Screen A branch.
- Add PMAI_APP_DB_PASSWORD to your deploy/.env. My copy is only in the worktree's .env, and I didn't change yours.
- Migrations now run separately, as the owner login. deploy/README.md has the new order: start the database, run migrations, then start the rest.
- The stack is still running at http://localhost:3000 if you want to try the screen. Sign in as admin with your seed admin password.

Limitations: end-to-end tests aren't in CI yet and run on Chromium only. No manual screen-reader check was done.

Next is your review of #3 and #2. Merging, and removing the two worktrees afterwards, needs your go-ahead.
```

Video 05:03. Built, caught, and still missing — separated.

AND WHAT THE HUMAN MUST DO

"Add `PMAI_APP_DB_PASSWORD` to your `deploy/.env` — my copy is only in the worktree's `.env` , and I didn't change yours." Then: "Merging, and removing the two worktrees afterwards, needs your go-ahead."

The mockup, now running

STEP
The
the mo

Home / Production orders / New

New production order

* Required fields are marked with an asterisk.

Order number

Assigned on save

Status

Draft

Product *

P-1001 — Steel bracket

Quantity *

11

Due date *

18/09/2026

Notes

Optional

0/500

Cancel

Save

Video 05:26. Localhost:3000/production-orders/new , signed in as Seed Admin.

The user squash-merges both pull requests from their own account (**DEC-030**) and opens the running application.

It is the same layout published as a private artifact in the detailed-design step — order number "Assigned on save", status Draft, the 30-product `<select>` , the notes counter, Cancel and Save — now backed by the API, the migration, the seeded database and the restricted `pmai_app` login.

THE GATE HELD

DEC-027 authorized branch, commit, push and open-PR. **Merge was never authorized for the agent**, and it never merged.

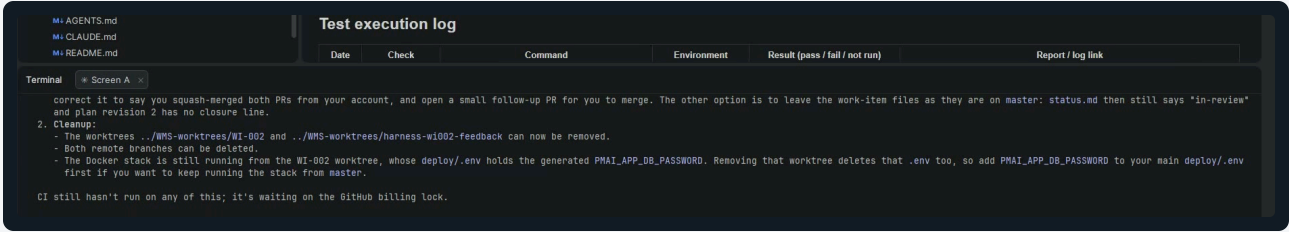
"The merge record is missing, and it's wrong"

After the squash merges, Claude checks what actually landed: "Your local `master` is now at `1eccf9c`, with your `demos/` files untouched. The only difference from PR #3's head is the 12 harness files that came in with #2, **so nothing was lost in the squash.**"

Then it reports a problem **with its own earlier work**: the commit that closes plan revision 2 exists only on the deleted branch, and its text says the merge used merge commits — which is not what happened.

TWO OPTIONS OFFERED, NOT ONE

Correct the record and open a small follow-up PR — or leave it, and accept that `status.md` still says "in-review" and revision 2 has no closure line.



Video 05:47. Cleanup listed too, including which `.env` disappears with the worktree.

RFC 0002 — keep every plan revision, oldest first

The second of the two improvements the Screen A series produced — and the step that also applied the first.

SCOPE 1 — THIS WORK ITEM

Revision 1 rebuilt

It had never been committed, so git could not restore it. Claude reconstructed it from the session's own edits, exactly as approved, and added a note saying so.

SCOPE 2 — THIS AGENT

The rule, then the file

"I'll update the saved rule first, then reorder the file." Every revision kept in full, oldest first, new revisions appended — the same ordering the detailed-design step used.

SCOPE 3 — EVERYONE

RFC 0002, after review

The user asked for the harness itself to be updated. `ai/templates/plan.md`, `ai/skills/planning/SKILL.md` and a new evaluation case.

ROOT CAUSE, AGAIN IN THE WORDING

The plan template said only "If this supersedes a prior revision, state which and why". **Nothing said to keep the old one.** The agent's reading was permitted; the rule was incomplete.

ALSO IN THIS SESSION

RFC 0001 was applied here — four templates, the templates README, the detailed-design skill and a new evaluation case — and committed on its own branch, because harness changes stay out of Screen A's pull request (DEC-028). Those opening minutes are covered by the detailed-design deck, not the cut.

ALSO OBSERVED HERE, FIXED IN RFC 0003

The implementation skill's branch-name example contradicted the project's branch policy, and the Playwright suite ran only locally. Both were closed later.

One screen, four steps, a full paper trail

STEP	AGENT TIME	PRODUCED
01 Basic design	6 m 16 s	Brief, decision log, BD-001
03 Database design	4 m 19 s	DB-002, DEC-013–DEC-020
02 Detailed design	10 m 23 s	4 DD documents + mockup
04 Implementation	~40 m	The screen, 133 tests, 2 PRs

Every requirement from REQ-010 to REQ-019 traces through a BD section, a DB constraint, a DD module, code and at least one test case — and `evidence.md` records each one as **verified**.

Sources: the session recording (`ScreenA_Implementaion.mp4`), its transcript, and the committed files in `work-items/WI-002/` and `ai/improvements/` .

WHAT THE HARNESS GAINED

- **RFC 0001** — always produce all four DD documents.
- **RFC 0002** — keep every plan revision, oldest first.
- Two evaluation cases, so neither regression returns unnoticed.

STILL OPEN AT THE END OF THE RECORDING

CI had not run — a GitHub billing lock. It first executed later, on PR #5, and now runs backend, frontend and E2E on every pull request.

Nine things to notice

Drawn from all four steps; the part each comes from is named.

01

Short prompts, structured harness

Thirteen prompts, most of them one line. `AGENTS.md` → workflows → skills → templates → checklists told the agent what to read and produce. *Part 1*

04

Asking again finds more

Twice a re-check by the user surfaced things the agent had silently settled — three timezone/cancel/seed gaps, and two missing design documents. *Parts 2 and 3*

07

Gates are written, not remembered

Publishing the mockup, approving the plan, merging the PRs — each was a row in a file, honoured without being restated in the prompt. *Parts 3 and 4*

02

Read before write

Instructions, templates and the existing code — including the running app's own CSS, so the mockup looked like the product. *Parts 1 and 3*

05

Design feeds back upstream

A missing quantity bound found while writing the API contract became a decision and a revision of the basic design, not a silent choice in code. *Part 3*

08

Proved, not asserted

The restricted database login was closed by attempting the forbidden delete and recording the refusal; a real focus bug was caught by a test written from the requirement. *Part 4*

03

Humans decide business behaviour

The agent recommends; the user chooses. Recommendations were overridden twice, and both overrides were recorded as decisions. *Parts 1 and 2*

06

Fix the rule before the artifact

Corrected twice, the agent wrote the preference to memory *first*, then produced the missing output. *Parts 3 and 4*

09

The harness improves itself

Two runs exposed ambiguities in `ai/` and produced RFCs against it — with the change to shared files offered, not applied. *Parts 3 and 4*